Mind the Touch Gap: Real-to-Sim Contact Alignment for Dexterous Pen Spinning

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Abstract: Pen spinning is fast and contact-rich, making simulation transfer difficult. PenSpin [1] fine-tuned a proprioceptive student on successful hardware replays. We ask whether a small replay budget can instead align the contact observations used during simulator training. Using Isaac Gym [2] and Project SuperDex [3], we propose to learn a source-to-target contact mapper, re-distill a contact-aware student, and test it against proprio-only, raw-contact, randomized-contact, calibrated-contact, and target-fine-tuning baselines. This sim-to-sim study tests the value of contact-observation alignment without claiming to correct dynamics or establish hardware transfer.

Keywords: Dexterous manipulation, contact alignment, sim-to-sim transfer

1 Introduction

Reinforcement learning in simulation has made dexterous in-hand skills possible without collecting large numbers of difficult demonstrations [4, 5, 6]. Pen spinning is an especially demanding case: the pen must pass through rapid, repeated contacts, and a small error in timing can end a rotation. PenSpin [1] trained a privileged oracle in simulation and distilled a proprioceptive student, but transferring the student directly to hardware remained difficult. Its solution was to replay the oracle's actions open-loop on the hand, retain successful trajectories, and fine-tune the student on those demonstrations.

This result shows the value of even a small amount of target experience, while leaving open how that experience might improve subsequent training in simulation. Prior work has used real experience to adapt simulation parameters [7, 8, 9], including recent tactile calibration through trajectory replay [10]. For pen spinning, the contact signal is a natural place to investigate: it changes rapidly during a spin and may differ across simulators even when the same actions are applied. We study this question in a controlled sim-to-sim setting, training in Isaac Gym [2] and evaluating in Project SuperDex [3]. This gives us access to target contact labels and repeatable evaluations, although it does not establish transfer to hardware.

We ask whether a small number of target replay attempts can align contact observations for policy training. We learn a mapping from source link-level contact, proprioception, and preceding actions to target contact, freeze it, and re-distill a student in Isaac Gym. PenSpin's original student used proprioception alone; we explicitly add a contact-aware student and retain the original as a baseline. We will compare zero-shot performance in SuperDex with raw-contact and proprio-only students, then with PenSpin-style target fine-tuning at matched replay budgets. State-matched replays will help identify contact errors caused by trajectory drift. The mapping changes a policy input, not contact dynamics.

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2 Plan

We will first reproduce PenSpin's oracle and proprio-only student, add a student with binary contact on corresponding Allegro links, and check that the hand, pen, controller, and contact readings behave sensibly in SuperDex. From matched initial states, we will replay the same oracle actions in Isaac Gym and SuperDex, then train a temporal mapper using source contact, proprioception, and *preceding* actions to predict target contact. After re-distillation with the mapper frozen, the student will run in SuperDex using native target contact (Figure 1). We will compare it with the original proprio-only student, raw contact, contact randomization, simple threshold-and-delay calibration, and PenSpin-style fine-tuning. All target-data methods receive the same number of attempted replays; we will also report how many of those attempts yield successful demonstrations for fine-tuning. Tests will share target objects and initial states, with results reported across replay budgets and friction/compliance settings. Contact prediction scores and replay-state divergence will help interpret policy performance.

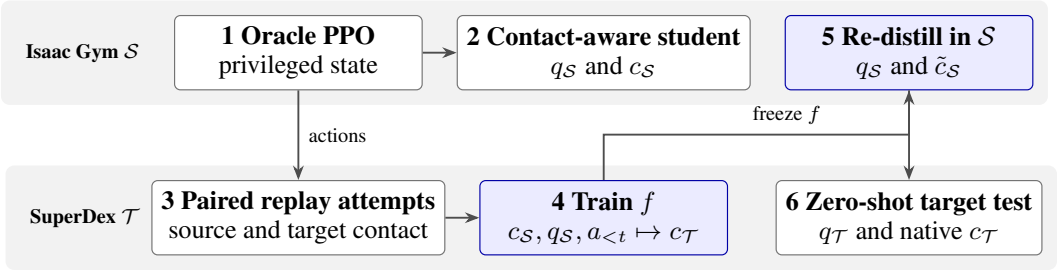


Figure 1: Proposed training and evaluation pipeline. The target simulator provides paired contact labels for the mapper and native contact for the final policy test.

- **10/13–10/30:** Reproduce PenSpin and complete a small Allegro/pen/contact replay test in SuperDex.
- **10/31–11/13:** Collect paired replays, train the contact mapper, and re-distill the student.
- **11/14–12/04:** Compare baselines at 10, 25, 50, and 100 replay attempts; measure rotations, drop time, success, contact F1, and state drift.
- **12/05–12/11:** Analyze results and prepare the final report and presentation.

If the SuperDex port proves unstable, we will first test a simpler in-hand rotation task [4] or move to a target simulator with explicit link-level contact, and report that change clearly.

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